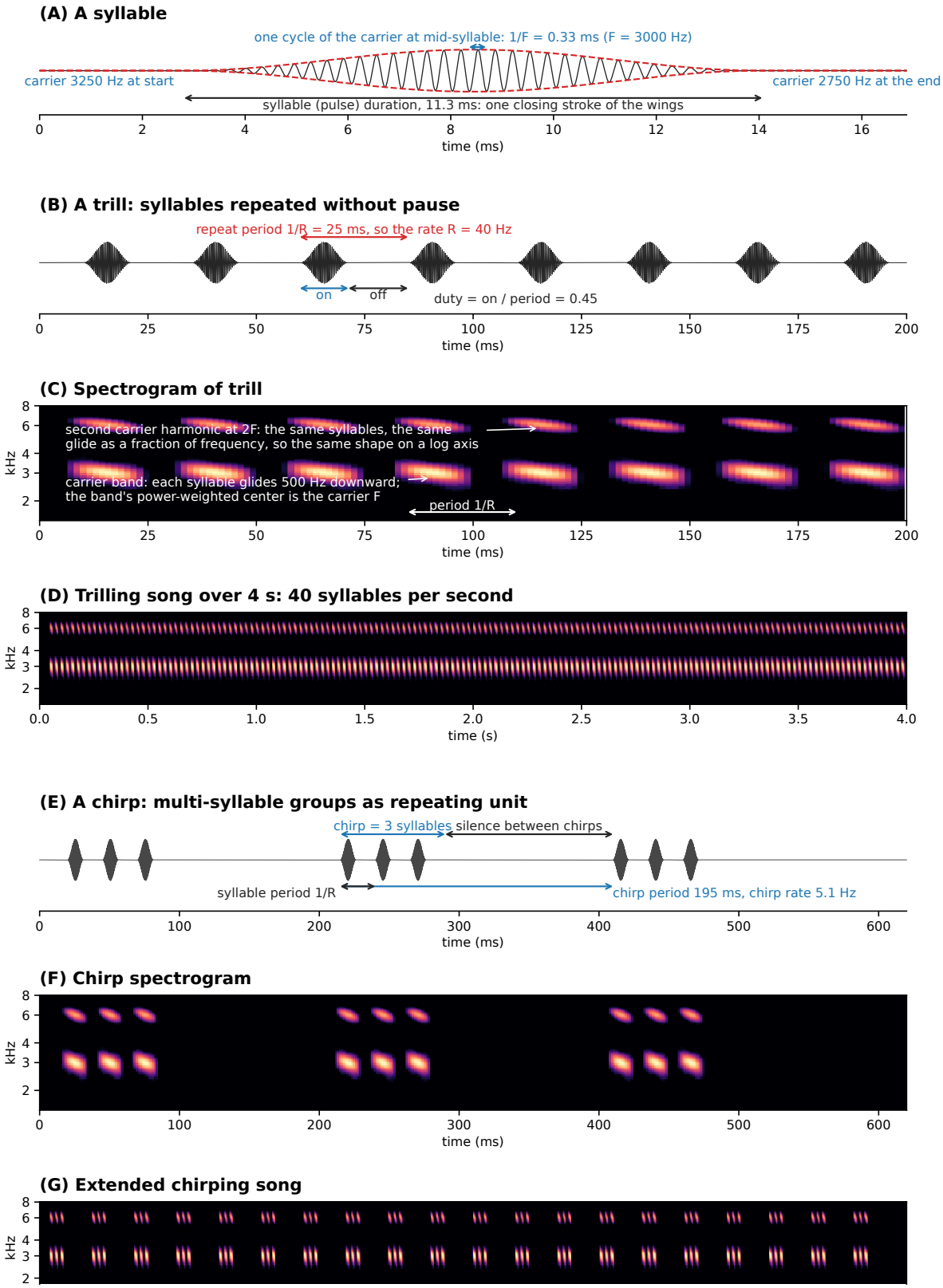

(A) A syllable
one cycle of the carrier at mid-syllable: 1/F = 0.33 ms (F = 3000 Hz)
carrier 3250 Hz at start
carrier 2750 Hz at the end
syllable (pulse) duration, 11.3 ms: one closing stroke of the wings
0 2 4 6 8 10 12 14 16
time (ms)
(B) A trill: syllables repeated without pause
repeat period 1/R = 25 ms, so the rate R = 40 Hz
on
off
duty = on / period = 0.45
0 25 50 75 100 125 150 175 200
time (ms)
(C) Spectrogram of trill
kHz
8 6 4 3 2
second carrier harmonic at 2F: the same syllables, the same glide as a fraction of frequency, so the same shape on a log axis
carrier band: each syllable glides 500 Hz downward; the band's power-weighted center is the carrier F
period 1/R
0 25 50 75 100 125 150 175 200
time (ms)
(D) Trilling song over 4 s: 40 syllables per second
kHz
8 6 4 3 2
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
time (s)
(E) A chirp: multi-syllable groups as repeating unit
chirp = 3 syllables
silence between chirps
syllable period 1/R
chirp period 195 ms, chirp rate 5.1 Hz
0 100 200 300 400 500 600
time (ms)
(F) Chirp spectrogram
kHz
8 6 4 3 2
0 100 200 300 400 500 600
time (ms)
(G) Extended chirping song
kHz
8 6 4 3 2

Figure 1.